

# 29.9% of returns aren't genuine. They carry **\$4.64M** of refund dollars — and now we know which ones.

A retailer's return-audit team reviews every disputed return the same way — spreadsheet speed, inconsistent criteria. We rebuilt **60,000 orders** into a PySpark/MLlib pipeline that triages each one into legitimate, policy abuse, fraud, or wardrobing, priced what's actually at stake, and separately segmented customers to surface the product-quality problem no fraud label can see.

## NON-LEGITIMATE REFUND EXPOSURE

# \$4.64M

**43.8%** of all refund dollars requested, from just **29.9%** of orders — bad actors skew toward higher-value claims

## CLASSIFIER PERFORMANCE (TEST SET)

# 0.999

weighted F1, Random Forest — synthetic-dataset caveat on slide 7, not a production guarantee

## HIDDEN IN THE "LEGITIMATE" MAJORITY

# 38%

of returns in the largest, 96%-legitimate customer segment are still quality-issue driven

**60,000 orders · 35 features**, Kaggle e-commerce return-abuse dataset · Fraudulent Return claims average **\$434.73** vs. **\$141.14** for legitimate returns — a **3.1x** gap · two independent PySpark MLlib approaches: multi-class classification (Random Forest vs. Logistic Regression) and K-Means clustering · this dataset is synthetic, so treat 0.999 F1 as a pipeline proof, not a production number (more on slide 7).

# One review queue, *three different problems* wearing the same form.

A return-audit team currently reviews every disputed return the same way. It has no systematic way to tell whether a spike in returns is fraud, abuse of a lenient policy, or a genuine quality defect — three problems that need three different owners, currently indistinguishable in a spreadsheet.

## 01 · CUSTOMERS RETURN

### Most returns are genuine

Defective items, wrong fit, changed mind — the normal cost of running an e-commerce store.

**70.1% legitimate**

## 02 · SOME ABUSE IT

### Policy abuse, fraud, wardrobing

Exploiting lenient terms, deceptive claims, or wearing-and-returning — three distinct behaviors.

**29.9% of orders**

## 03 · ALL LOOK THE SAME

### One manual-review process

Every disputed return gets identical treatment today, regardless of which of the three it actually is.

**1 review queue**

## 04 · THE WRONG FIX

### Blanket policy tightening

Punishes the 70% of legitimate customers to catch the fraction who abuse the system.

**wrong lever**

## THE QUESTION THIS PROJECT ANSWERS

*Which returns are fraud, which are policy abuse, and which are genuine quality failures — and what does each one actually cost?*

Three sub-problems, one dataset: **return fraud** → classification on abuse\_type; **product quality** → return-reason patterns cut by cluster and category; **return risk** → lifetime order/return history per customer.

## WHY THIS IS A TRIAGE PROBLEM, NOT A POLICY PROBLEM

The highest-value lever is routing each return to the right team fast — not tightening policy across the board, which would punish the 70% of customers who did nothing wrong. **Three different interventions, three different owners**, is the recommendation this whole pipeline builds toward.

## INDUSTRY BACKGROUND

# This is not a hypothetical problem — *the industry is already living it.*

E-commerce carries a structurally higher return burden than brick-and-mortar, and fraud/abuse is a large, growing share of it. This project sits squarely inside an industry-wide shift already underway.

### NRF · 2025 RETAIL RETURNS LANDSCAPE

# 19.3%

of online sales are returned — above the 15.8% all-channel average. \$849.9B in merchandise returns forecast for 2025.

### APPRISS RETAIL × DELOITTE · 2024

# 15.14%

of all returns are fraudulent — \$103B lost in 2024 (up from \$101B in 2023), across 60+ retailers studied.

### SAME 2024 STUDY

# 60%

of retail executives specifically flag **wardrobing** — one of our four target classes — as a major fraud pattern. 85% now deploy AI to fight it.

### HOW OUR DATASET COMPARES

This dataset's combined Fraudulent Return + Wardrobing rate is **10.2% + 7.7% = 17.9%** — close to Appriss's industry-wide **15.14%** fraud rate. The class **prevalence** here is realistically calibrated to real-world reporting, even though (see slide 7) the class **separability** the model achieves is not something to expect on live data.

Sources: National Retail Federation, *2025 Retail Returns Landscape* (nrf.com) · Appriss Retail with Deloitte, *Fraudulent Returns and Claims Cost Retailers \$103B in 2024* (apprissretail.com). Full citations in the report's Appendix C.

# A realistic imbalance — and the dollars don't split the way the order counts do.



■ Legitimate — 70.1% (42,060 orders, \$5.94M requested) ■ Policy Abuser — 12.0% (7,192, \$0.76M) ■ Fraudulent Return — 10.2% (6,112, \$2.66M) ■ Wardrobing — 7.7% (4,636, \$1.22M)

## Behavioral signals

Photo evidence, tracking validity, address changes, refund-to-different-account, multiple accounts, dispute history

## Customer history

Lifetime orders/returns, return rate, account age, loyalty segment

## Context

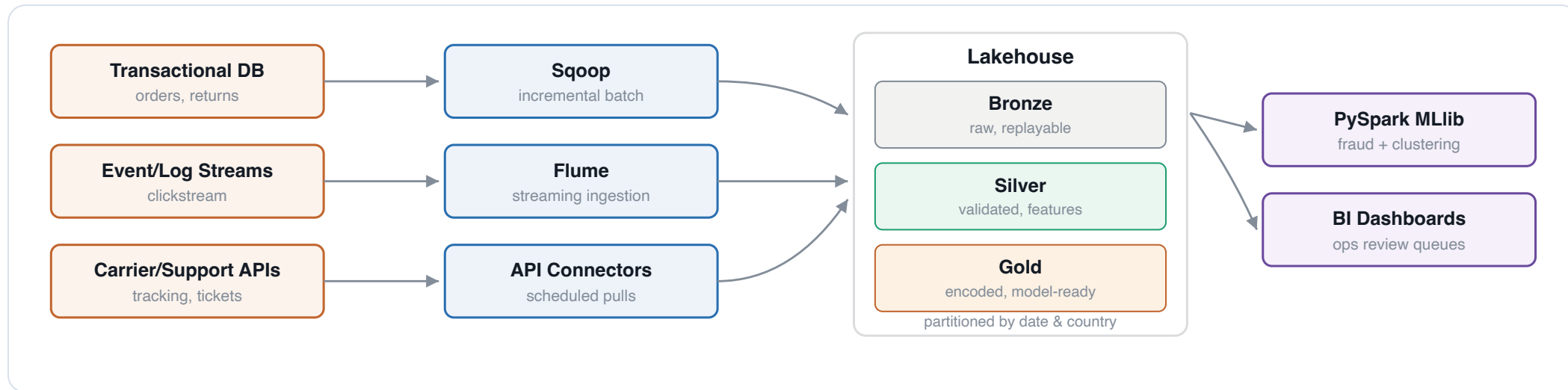
12 product categories, 12 return reasons, 8 countries, payment method, shipping carrier

## VERIFIED — CHECKED PROGRAMMATICALLY, NOT ASSUMED

**0 missing values** across all 35 columns, **0 duplicate** order\_id, **0 logical violations** (no return-before-order dates, no returns exceeding lifetime orders, no implausible ages) — confirmed in notebooks/01\_data\_engineering.ipynb §2 before anything downstream trusted this data.

# Sqoop / Flume / API ingestion → *Bronze/Silver/Gold lakehouse* → PySpark MLlib.

This notebook runs directly on the static Kaggle CSV as a practical stand-in — the diagram below is the production-scale design the Silver→Gold transformation logic is built to generalize to.



## WHY BRONZE/SILVER/GOLD

Bronze keeps a permanent, replayable raw record. Silver is cleaned and feature-engineered once, reused by both models. Gold is the only layer anything ML touches.

## EXPLICIT SCHEMA, NOT INFERRED

The notebook loads the CSV with an explicit StructType — never inferSchema in production, which forces a full extra scan and can silently mis-type columns on drift.

## CAP OUTLIERS, DON'T DELETE

Extreme values are often the fraud signal itself. 8 continuous fields IQR-capped at the 1st/99th percentile — preserving row count and the extremeness of the behavior.

# Engineered features *validated* before they ever reach a model.

| FEATURE                                                                                                                      | SERVES          |
|------------------------------------------------------------------------------------------------------------------------------|-----------------|
| fraud_signal_score                                                                                                           | Return fraud    |
| return_to_order_ratio                                                                                                        | Return risk     |
| refund_to_order_value_ratio                                                                                                  | Return fraud    |
| is_quality_issue_reason                                                                                                      | Product quality |
| high_value_no_evidence                                                                                                       | Return fraud    |
| Outliers capped at 1st/99th percentile, not deleted — extreme values are often the fraud signal itself, not noise to remove. |                 |

| VERIFIED — PRINTED BY THE RUN, NOT CLAIMED                                                                                                                                                                                                                                   |                   |            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|
| Mean <b>fraud_signal_score</b> : 2.73 for Fraudulent Return vs. 0.84 for Legitimate. Mean <b>return_to_order_ratio</b> : 0.47 vs. 0.05. Clear separation confirmed <i>before</i> trusting these features for modeling — not assumed. notebooks/01_data_engineering.ipynb §4. |                   |            |
|                                                                                                                                                                                                                                                                              | FRAUDULENT RETURN | LEGITIMATE |
| fraud_signal_score                                                                                                                                                                                                                                                           | 2.73              | 0.84       |
| return_to_order_ratio                                                                                                                                                                                                                                                        | 0.47              | 0.05       |

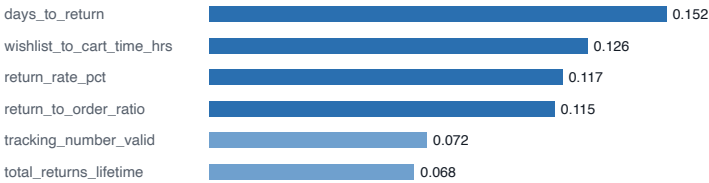
# Random Forest vs. Logistic Regression on *return-fraud type*.

Spark MLlib's `GBTClassifier` only supports binary classification — not usable for this 4-class target — so Random Forest, which natively supports multi-class, is the primary model. Class-weighted training (inverse frequency) handles the 70/12/10/8 imbalance; evaluation reports per-class precision/recall/F1, not headline accuracy.

| MODEL                          | WEIGHTED PRECISION | WEIGHTED RECALL | WEIGHTED F1 |
|--------------------------------|--------------------|-----------------|-------------|
| Logistic Regression (baseline) | 0.998              | 0.998           | 0.998       |
| Random Forest (final)          | 0.999              | 0.999           | 0.999       |

200 trees, maxDepth=10, class-weighted. Train 45,112 / test 14,888 rows, 75/25 split, seed=42.

## TOP FEATURE IMPORTANCES



Engineered ratios rank alongside raw behavioral fields — the payoff of Phase 2.

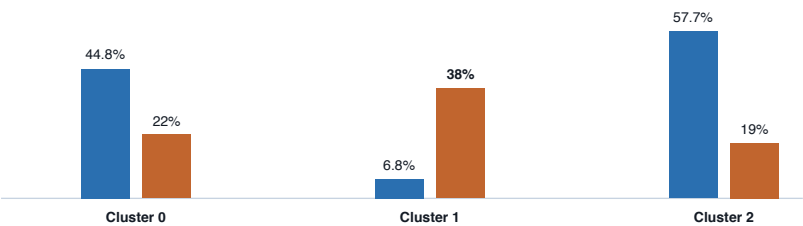
**Honest note:** this is a Kaggle synthetic dataset, so 0.999 F1 is higher than we'd expect in production — feature importance is spread across ~15 features with no single leak, but a realistic target on live data is **80–90% F1**. We treat this model as proof of the pipeline, not a performance guarantee. Full discussion in `docs/fraud-classification-results.md`.

APPROACH 2 — CLUSTERING

# K-Means finds 3 behavioral segments *the fraud label can't see.*

An independent, unsupervised lens: features deliberately exclude the fraud-flag columns used by the classifier, so clustering surfaces behavior/quality patterns a supervised label alone can't. k=3 selected by silhouette score over k=2..6.

| CLUSTER                   | N      | AVG RETURN RATE | AVG ORDER VALUE | % QUALITY-ISSUEDOMINANT MIX |
|---------------------------|--------|-----------------|-----------------|-----------------------------|
| 0 — High-value fraud-risk | 6,646  | 44.8%           | \$484           | 22% Fraudulent Return-heavy |
| 1 — Core legitimate base  | 43,829 | 6.8%            | \$157           | 38% 96% Legitimate          |
| 2 — Frequent policy-abuse | 9,525  | 57.7%           | \$129           | 19% Policy Abuser-heavy     |



KEY INSIGHT

Cluster 1 has by far the **lowest** return rate but the **highest** quality-issue share — the two metrics move in opposite directions, which is exactly why a single blended "return rate" KPI hides this segment's problem from whoever owns it today.

This is a **QA/vendor problem**, not a fraud problem — cross-reference dominant product\_category within Cluster 1 against return\_reason to target the escalation.

# Four interventions, four owners — *ship the free one first.*

## SHIP THIS FIRST, NO ML REQUIRED

Require photo evidence above a value threshold. The "high-value item + no photo evidence" pattern alone touches **\$1.88M** of non-legitimate refund exposure across **4,673 orders** — a policy rule, live immediately, independent of the classifier's rollout timeline.

### 01 Deploy the fraud classifier as a triage score, not an auto-block.

Route the highest `fraud_signal_score` decile + Cluster 0 to manual review; auto-approve the lowest-risk tail. **Re-threshold asymmetrically** — a false "Legitimate → Fraudulent Return" flag costs more in trust than a slow review costs in ops time.

**OWNER · FRAUD & RISK**

### 02 Give Cluster 1's 38% quality-issue rate to QA as a systemic audit, not a vendor hunt.

Quality-issue rates are broad-based — banded 31.5%–35.2% across all 12 categories, not one bad vendor. **73% of customers**, near-zero fraud, but the highest quality complaint rate of any segment.

**OWNER · QA / VENDOR MANAGEMENT**

### 03 Tighten policy — don't investigate — for Cluster 2.

High order/return volume, low order value, Policy-Abuser-dominated. A shorter return window or restocking fee targets this pattern more precisely than a fraud investigation would.

**OWNER · CUSTOMER POLICY**

### 04 Feed the refund-exposure regressor to Finance, not just Fraud & Risk.

Linear Regression prices expected refund dollars the moment a return is filed — a reserve-accounting input independent of whether it turns out to be fraud, abuse, or genuine.

**OWNER · FINANCE / CUSTOMER OPS**

**Risks & roadmap:** re-validate on real labeled outcomes before trusting this dataset's metrics; no temporal holdout here, so add a rolling time-based validation window in production; build the human-in-the-loop labeling loop from the review queue this system creates.



THANK YOU

# Thank you. Questions *welcome.*

We'd rather be asked about the caveats than the number —  
**\$4.64M** of non-legitimate refund exposure, **\$1.88M** of it  
reachable today with a single no-ML policy rule.

Subham Ranjan · A017

Shriya · A023

Abhinav Kumar · A035

Dev Vagreacha · A049

Samruddhi Pradhan · A054

Yatharth Vij · A072

## IN THE REPOSITORY

notebooks/ — two executable  
PySpark notebooks, Bronze to  
models

## WRITTEN UP

docs/ — per-model results, each  
with its own weaknesses section

## THE DELIVERABLES

deliverables/ — this deck and the  
full consulting report, HTML + PDF

## REPRODUCIBILITY

seed=42 throughout · every number  
here read from the notebooks' own  
output

Dataset: `ecommerce_return_abuse_dataset.csv` (Kaggle) · references, AI-usage disclosure and team contributions in the consulting report's Appendix.